



# CAT 2025

MBA RAPID REVISION

Lecture - 02

Revision Class - 2 ✓

Routes & Networks  
Games & Tournaments ↗ ★

By- GOURAV GUPTA



# TOPICS *to be covered*

1. Rapid Revision

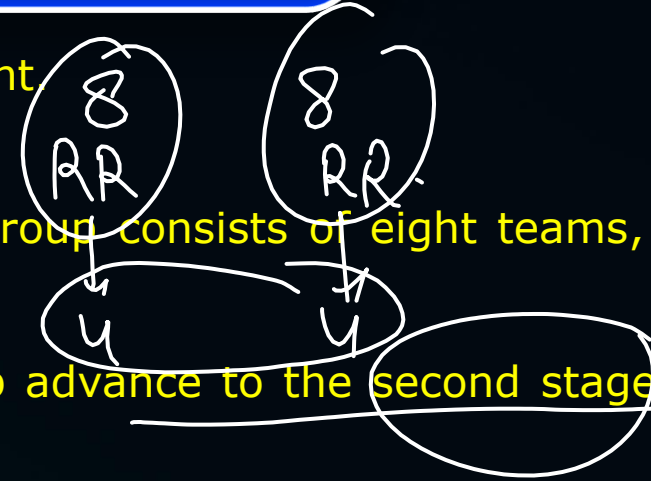


**Set-1**

## Directions :



- Sixteen teams have been invited to participate in cricket tournament
- The tournament is conducted in two stages.
- In the first stage, the teams are divided into two groups. Each group consists of eight teams, with each team playing every other team in its group exactly once.
- At the end of the first stage, the top four teams from each group advance to the second stage while the rest are eliminated. The second stage comprises of several rounds.
- A round involves one match for each team.
- The winner of a match in a round advances to the next round, while the loser is eliminated. The team that remains undefeated in the second stage is declared the winner.
- The tournament rules are such that each match results in a winner and a loser with no possibility of a tie.
- In the first stage, a team earns one point for each win and no points for a loss.
- At the end of the first stage, teams in each group are ranked on the basis of total points to determine the qualifiers advancing to the next stage.
- Ties are resolved by a series of complex tie-breaking rules so that exactly four teams from each group advance to the next stage.



#Q. What is the total number of matches played in the tournament? 63.

#Q. The minimum number of wins needed for a team in the first stage to guarantee its advancement to the next stage is: 6

#Q. What is the highest number of wins for a team in the first stage in spite of which it might be eliminated at the end of first stage? 5

#Q. What is the number of rounds in the second stage of the tournament? 3

## Analysis :

$$8 = 2^n = 2^3$$



Stage-1 (Round Robin)

$8 \rightarrow 28$   
 $8 \rightarrow 28$  } 56 matches

Stage-2 (Knock-out)

$4 \rightarrow$   
 $+$   
 $4 \rightarrow$  } 8 Teams

Round = 3

Matches =  $8 - 1 = 7$

- Sixteen teams have been invited to participate in cricket tournament.
- The tournament is conducted in two stages.
- In the first stage, the teams are divided into two groups. Each group consists of eight teams, with each team playing every other team in its group exactly once.
- At the end of the first stage, the top four teams from each group advance to the second stage while the rest are eliminated. The second stage comprises of several rounds.
- A round involves one match for each team.
- The winner of a match in a round advances to the next round, while the loser is eliminated. The team that remains undefeated in the second stage is declared the winner.
- The tournament rules are such that each match results in a winner and a loser with no possibility of a tie.
- In the first stage, a team earns one point for each win and no points for a loss.
- At the end of the first stage, teams in each group are ranked on the basis of total points to determine the qualifiers advancing to the next stage.
- Ties are resolved by a series of complex tie-breaking rules so that exactly four teams from each group advance to the next stage.

# Analysis :

8 Teams.

$$\text{Matches} = {}^nC_2 = \frac{n(n-1)}{2}$$

$$= {}^8C_2 = \textcircled{28}$$

A → 7 5

B → 6 5

C → 5 5

D → 4 5

E → 3 5

F → 2 2

G → 1 1

H → 0 0

① Min Win → Guarantee

⑥

→ (5+1) ✓

3 out of 28

$$\text{Rem} = \frac{\textcircled{25}}{5} = \textcircled{5}$$

Top 4

10 Teams.  $\rightarrow$  45 matches.



Top 4  $\rightarrow$  Qualify

min Guarantee  $\rightarrow (7+1) = \textcircled{8}$   
wins

Max Win (Not qualify) =  $\textcircled{7}$

min win (may qualify) =

|       |                 |
|-------|-----------------|
| A     | $\rightarrow 7$ |
| B     | $\rightarrow 7$ |
| C     | $\rightarrow 7$ |
| D     | $\rightarrow 7$ |
| <hr/> |                 |
| E     | $\rightarrow 7$ |
| F     | $\rightarrow 4$ |
| G     | $\rightarrow 3$ |
| H     | $\rightarrow 2$ |
| I     | $\rightarrow 1$ |
| J     | $\rightarrow 0$ |

10 wins.

$$\text{Rem} = 45 - 10 = \textcircled{35}$$



8 Teams  $\rightarrow$  28 matches.

$$28 - 18 = (10)$$

|       |                 |   |
|-------|-----------------|---|
| A     | $\rightarrow$ 7 | 7 |
| B     | $\rightarrow$ 6 | 6 |
| C     | $\rightarrow$ 5 | 5 |
| D     | $\rightarrow$ 4 | 2 |
| <hr/> |                 |   |
| E     | $\rightarrow$ 3 | 2 |
| F     | $\rightarrow$ 2 | 2 |
| G     | $\rightarrow$ 1 | 2 |
| H     | $\rightarrow$ 0 | 2 |

10 Teams  $\rightarrow$  45

|       |                 |
|-------|-----------------|
| A     | $\rightarrow$ 9 |
| B     | $\rightarrow$ 8 |
| C     | $\rightarrow$ 7 |
| D     |                 |
| <hr/> |                 |
| E     |                 |
|       |                 |
|       |                 |
|       |                 |
|       |                 |
|       |                 |
|       |                 |
|       |                 |
|       |                 |

$\frac{21}{7} = (3)$



11 Teams  $\rightarrow$  55 match.

Top 5  $\rightarrow$  Qualify

$$\left. \begin{array}{l} A \rightarrow 10 \\ B \rightarrow 9 \\ C \rightarrow 8 \\ D \rightarrow 7 \end{array} \right\} = 34$$

E

F

1

1

1

1

1

1

$$\text{Rem} = \frac{21}{7} = 3$$



11 Teams  $\rightarrow$  55 matches.

Top 4  $\rightarrow$  Qualify

Min Wins (May Qualify) = 4

min wins (Guarantee) = 9.

max win (may not qualify) = 8

|       |               |    |    |
|-------|---------------|----|----|
| A     | $\rightarrow$ | 10 | .. |
| B     | $\rightarrow$ | 9  | .. |
| C     | $\rightarrow$ | 8  | x  |
| D     | $\rightarrow$ | 7  | x  |
| <hr/> |               |    |    |
| E     | $\rightarrow$ | 6  | .. |
| F     | $\rightarrow$ | 5  | .. |
| G     | $\rightarrow$ | 4  | .. |
| H     | $\rightarrow$ | 3  | .. |
| I     | $\rightarrow$ | 2  | .. |
| J     | $\rightarrow$ | 1  | .. |
| K     | $\rightarrow$ | 0  | .. |

$$\text{Rem} = 55 - 27 = \frac{28}{8} = 3.5$$

3 or 4

3 win x



**Set-2**

Read the following data and answer the given questions.



In the All-Indian soccer finals, 6 teams qualified: Kerala Blasters, Mohan Bagan, Goa, Bengaluru, Hyderabad and Mumbai. In the first round, each team plays the other team once. Each win earns a team three points, a draw earns one point, and a loss earns zero points. Next, the top 4 teams move to the semi-finals where the first team in the first round plays the fourth and the second team plays the third. In case of a tie, the team with the highest goal difference (Goal for – Goal against) qualifies. The winners of the semi-finals compete in the finals.

| Teams           | Games Played | Won | Drawn | Lost | Goals For | Goals Against |
|-----------------|--------------|-----|-------|------|-----------|---------------|
| Goa             | 3            | 3   | 0     | 0    | 7         | 1             |
| Mohan Bagan     | 3            | 2   | 0     | 1    | 2         | 1             |
| Hyderabad       | 3            | 1   | 1     | 1    | 4         | 2             |
| Mumbai          | 4            | 1   | 1     | 2    | 4         | 3             |
| Kerala Blasters | 3            | 0   | 3     | 0    | 3         | 3             |
| Bengaluru       | 4            | 0   | 1     | 3    | 2         | 12            |

In the opening match of the tournament, Hyderabad loses to Mohan Bagan by 1-0. Please note that none of the matches in the first round were goalless draws.

## Questions



#Q-1. Which of the following teams has Mohan Bagan not played against?

- A. Goa      ✓ B. Bengaluru      C. Hyderabad      D. Mumbai

#Q-2. What was the highest goal difference in any game?

- A. 3 goals      ✓ B. 4 goals      C. 5 goals      D. 6 goals

$$5 - 1 = 4$$

#Q-3. How many teams are still contenders for a semi-final berth?

- A. 3      B. 4      C. 5      ✓ D. 6

#Q-4. If both Goa and Mohan Bagan won their remaining matches, then which of the following can be a semi-final fixture?

- A. Hyderabad vs Mumbai ✗      ✓ B. Goa vs Hyderabad  
C. Mohan Bagan vs Kerala Blasters ✗      ✗ D. Goa vs Mohan Bagan

Goa → Hyd | Mum  
MB → Mum | Hyd

## Analysis



✓ KB - Hy (1-1)    ✓ MB - Hy (1-0)  
 ✓ KB - Mum (1-1)    ✓ Hyd - Beng (3-0)  
 ✓ KB - Beng. (1-1)    ✓ MB - Mum (1-0)  
                                  ✓ Mum - Beng. (3-0)

✓ Goa - MB (1-0)  
 ✓ Goa - Mum (1-0)  
 Goa - Beng. (5-1)

G → 15  
 MB → 12  
 Hy → 4 +  
 Mum → 4 +  
 KB → 3  
 Beng. → 1

$G_2 = 15$

KB - Goa  
 KB - MB  
 Goa - Hyd  
 MB - Beng  
 Hyd - Mum

| Teams           | Points | Games Played | Won   | Drawn | Lost | Goals For | Goals Against |
|-----------------|--------|--------------|-------|-------|------|-----------|---------------|
| Goa             | 9      | 111 3        | 111 3 | 0     | 0    | 115 7     | 1 1           |
| Mohan Bagan     | 6      | 3111         | 211   | 0     | 1 1  | 211       | 1 1           |
| Hyderabad       | 4      | 3111         | 1 1   | 1 1   | 1 1  | 413       | 11 2          |
| Mumbai          | 4      | 4111         | 1     | 1 1   | 211  | 413       | 11 3          |
| Kerala Blasters | 3      | 3 3          | 0     | 3 3   | 0    | 3 3       | 3 3           |
| Bengaluru       | 1      | 4111         | 0     | 1 1   | 311  | 121       | 13 12 3       |

In the opening match of the tournament, Hyderabad loses to Mohan Bagan by 1-0. Please note that none of the matches in the first round were goalless draws.



## Set-3



Read the following data and answer the given questions.



In an 8×8 chessboard, a queen placed anywhere can attack another piece if the piece is present in the same row, in the same column, or any diagonal position in any possible 4 directions, provided there is no other piece in between in the path from the queen to that piece.

The columns are labelled a to h (left to right), and the rows are numbered 1 to 8 (bottom to top). The position of a piece is given by the combination of column and row labels. For example, position c5 means that the piece is in Column C and Row 5.



## Questions



#Q-1. If the queen is at c5, and the other pieces at positions c2, g1, g3, g5, and a3, then how many are under attack by the queen? There are no other pieces on the board.

- A. 2                      B. 3                      C. 4                      D. 5

#Q-2. If the other pieces are only at positions a1, a3, b4, d7, h7, and h8, then which of the following positions of the queen results in the maximum number of pieces being under attack?

- A. f8                      B. a7                      C. c1                      D. d3

#Q-3. Suppose the queen is the only piece on the board and it is at position d5. In how many positions can another piece be placed on the board such that it is safe from attack from the queen?

- A. 32                      B. 36                      C. 35                      D. 37

#Q-4. If the other pieces are only at positions a1, a3, b4, d7, h7, and h8, then from how many positions the queen cannot attack any of the pieces?

- A. 0                      B. 3                      C. 4                      D. 6

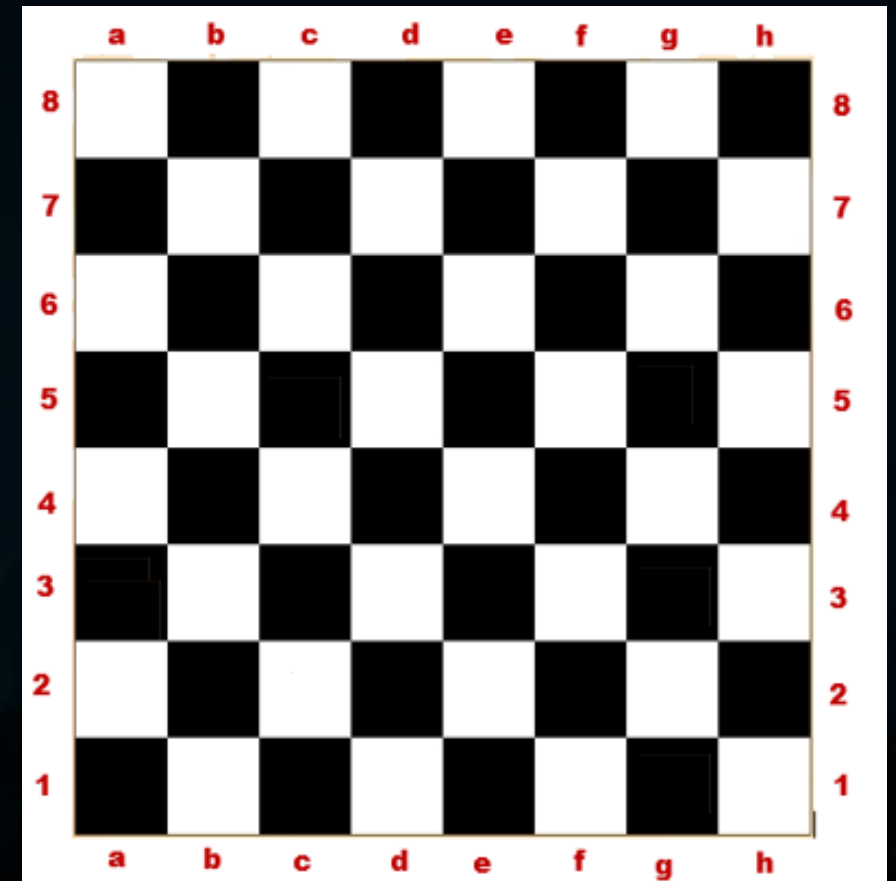
Read the following data and answer the given questions.



In an 8×8 chessboard, a queen placed anywhere can attack another piece if the piece is present in the same row, in the same column, or any diagonal position in any possible 4 directions, provided there is no other piece in between in the path from the queen to that piece.

The columns are labelled **a to h (left to right)**, and the rows are numbered **1 to 8 (bottom to top)**. The position of a piece is given by the combination of column and row labels.

For example, position c5 means that the piece is in Column C and Row 5.

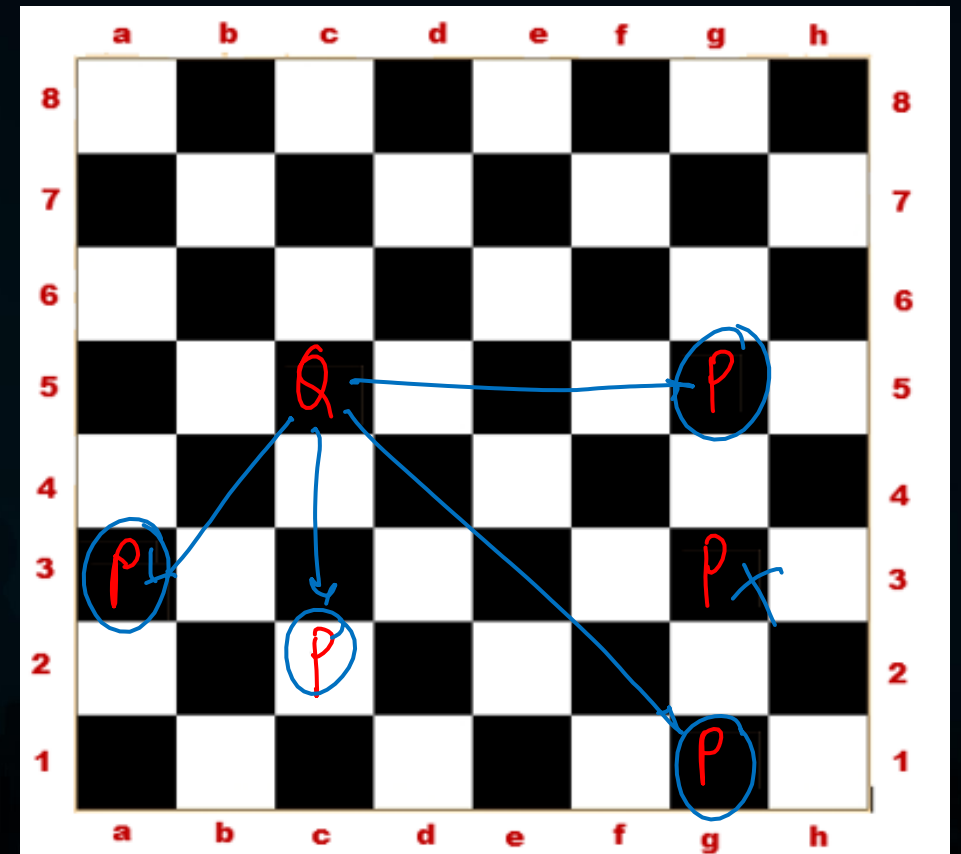


## Questions



#Q-1. If the queen is at c5, and the other pieces at positions c2, g1, g3, g5, and a3, then how many are under attack by the queen? There are no other pieces on the board.

- A. 2      B. 3      C. 4      D. 5



## Questions



#Q-2. If the other pieces are only at positions a1, a3, b4, d7, h7, and h8, then which of the following positions of the queen results in the maximum number of pieces being under attack?

A. f8

2

B. a7

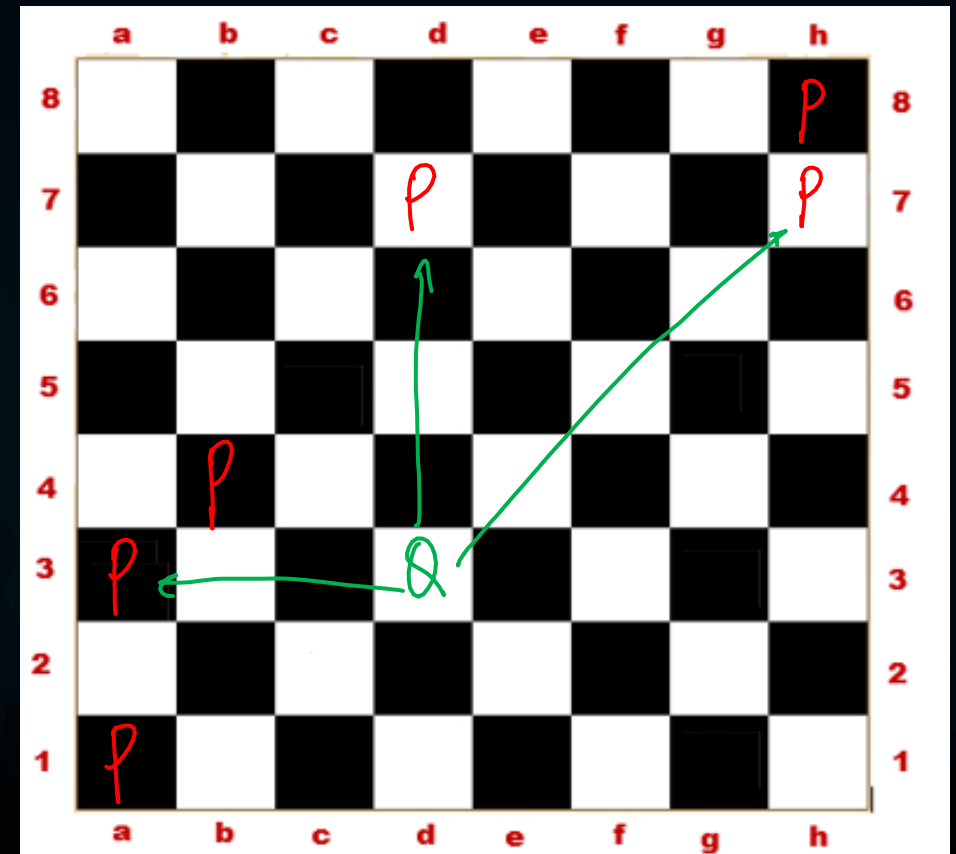
2

C. c1

2

~~D. d3~~

3



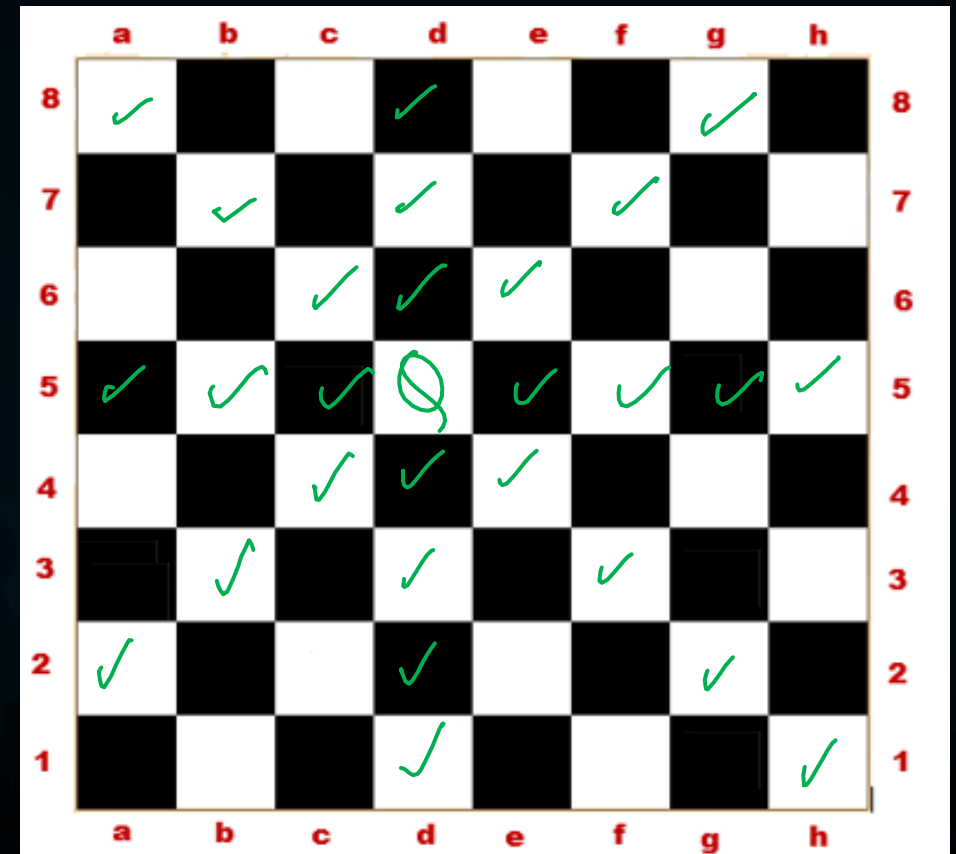
## Questions



#Q-3. Suppose the queen is the only piece on the board and it is at position d5. In how many positions can another piece be placed on the board such that it is safe from attack from the queen?

- A. 32      ☒ B. 36      C. 35      D. 37

$$27 + 1$$
$$64 - 28 = \textcircled{36}$$



## Questions



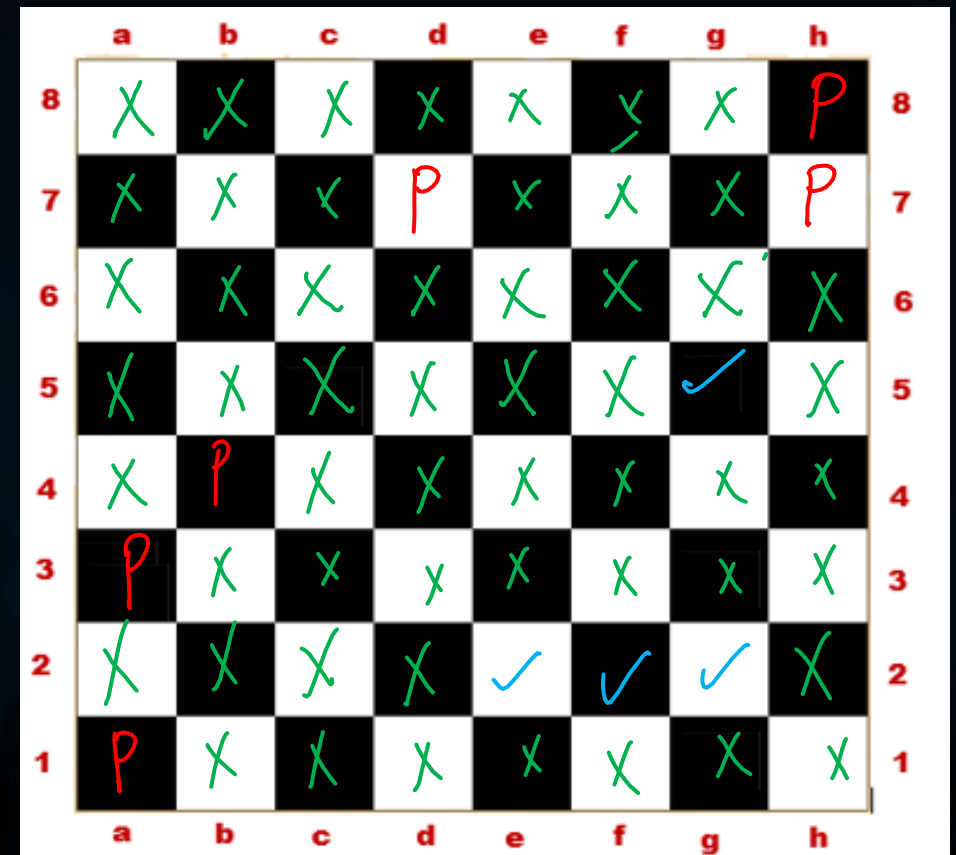
#Q-4. If the other pieces are only at positions a1, a3, b4, d7, h7, and h8, then from how many positions the queen cannot attack any of the pieces?

A. 0

B. 3

✓ C. 4

D. 6





Thank You.







THANK  
*You*

